
EECS 16B Designing Information Devices and Systems II

Discussion Worksheet

Discussion 10A

1. Towards Upper-Triangulation By An Orthonormal Basis

Previously in this course, we have seen the value of changing our coordinates to be eigenbasis-aligned, because we can then view the system as a cascade of scalar systems. If we have a diagonalization, then these scalar equations are fully uncoupled, and can therefore be treated completely separately. But even when we cannot diagonalize, we can upper-triangularize in a way that allows us to solve the equations one at a time, from the "bottom up".

In this problem, to better understand the steps involved, we will use the following concrete example:

$$S_{[3 \times 3]} = \begin{bmatrix} \frac{5}{12} & \frac{5}{12} & \frac{1}{6} \\ \frac{5}{12} & \frac{5}{12} & \frac{1}{6} \\ \frac{1}{6} & \frac{1}{6} & \frac{2}{3} \end{bmatrix} \quad (1)$$

and figure out the general case by abstracting variables. Note that there is a datahub link to a jupyter notebook on the website, which will allow you to perform the numerical calculations quickly to connect the symbolic analysis to an example, but without being time-consuming in the process.¹

- (a) Consider a non-zero vector $\vec{u}_0 \in \mathbb{R}^n$. Can you think of a way to extend it to a set of basis vectors for $\mathbb{R}^n$? In other words, find $\vec{u}_1, \dots, \vec{u}_{n-1}$, such that $\text{span}(\vec{u}_0, \vec{u}_1, \dots, \vec{u}_{n-1}) = \mathbb{R}^n$. To begin with, consider $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Can you get an orthonormal basis from what you just constructed?

Hint: what was the last discussion all about? Also, the given vector isn't normalized yet!

¹This particular matrix has an additional special property of symmetry, but we won't be invoking that here.

- (b) Now consider a real eigenvalue λ_1 , and the corresponding (normalized) eigenvector $\vec{v}_1 \in \mathbb{R}^n$ of a square matrix $M \in \mathbb{R}^{n \times n}$. From the previous part, we know that we can extend $\vec{v}_1$ to an orthonormal basis of $\mathbb{R}^n$. We will denote the basis by

$$U = \begin{bmatrix} | & | & \cdots & | \\ \vec{u}_1 & \vec{u}_2 & \cdots & \vec{u}_n \\ | & | & \cdots & | \end{bmatrix}$$

where $\vec{u}_1 = \vec{v}_1$ (note that this eigenvector is already normalized).

Our goal is to look at what the matrix M looks like in the coordinate system defined by the basis U .

Compute $U^\top M U$ by writing $U = [\vec{v}_1 \quad R]$, where $R \triangleq \begin{bmatrix} | & | & \cdots & | \\ \vec{r}_1 & \vec{r}_2 & \cdots & \vec{r}_{n-1} \\ | & | & \cdots & | \end{bmatrix}$.

- (c) Show that $U^{-1} = U^\top$.

(d) Define $Q = R^\top MR$. Look at the first column and the first row of $U^\top MU$ and show that:

$$M = U \begin{bmatrix} \lambda_1 & \vec{a}^\top \\ \vec{0} & Q \end{bmatrix} U^\top$$

Here, $\vec{a}$ is a symbolic vector related to M , R , and $\vec{v}_1$ (we will show the relation!).

(e) Now, we can recurse on Q to get:

$$Q = \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix} \begin{bmatrix} \lambda_2 & \vec{b}^\top \\ \vec{0} & P \end{bmatrix} \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}^\top$$

where we have taken $\vec{v}_2 \in \mathbb{R}^{n-1}$, a normalized eigenvector of Q , associated with eigenvalue λ_2 . Again $\vec{v}_2$ is extended into an orthonormal basis to form $\begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}$.

Plug this form of Q into M above, to show that:

$$M = \begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix} \begin{bmatrix} \lambda_1 & \check{a}_1 & \check{a}_{\text{rest}}^\top \\ 0 & \lambda_2 & \vec{b}^\top \\ \vec{0} & 0 & P \end{bmatrix} \begin{bmatrix} \vec{v}_1 & R\vec{v}_2 & RY \end{bmatrix}^\top$$

where we define $\check{a}$ to be the "adjusted" $\vec{a}$ to account for the substitution of Q ; $\check{a}^\top = \vec{a}^\top \begin{bmatrix} \vec{v}_2 & Y \end{bmatrix}$.

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